VoiceOps 2026 Capstone Report: ZentroMart AI Support Agent

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LLM & Generative AI Course (C&A Club, IIT Guwahati x Gnani.ai) | Reference Date Used: 20 October 2026

1. What We Built in Each Step

Our goal in this project was to build a customer support agent for ZentroMart (a fictional Indian online store) that answers customer questions from company documents, checks order records safely, and talks to callers over voice using Gnani's speech APIs. We built and tested the system in four steps:

- **Step 1 (Prompt-Only LLM Baseline):** We used Qwen/Qwen2.5-3B-Instruct on a T4 GPU in Google Colab. We gave it a simple system prompt telling it to act as ZentroMart's support assistant and start every reply with 'You are talking to an AI assistant: '. Because it had no access to ZentroMart's policy files or product catalog in this step, it had to guess policies, warranty periods, and refund timelines.
- **Step 2 (Grounded RAG over Company Documents):** We loaded the 20 active Markdown files from the kb/ folder (skipping the meta file 00_about_this_knowledge_base.md) and split them by Markdown headings (## and ###) into 100 section chunks. Each heading in the dataset has a section code like [RET-2] (the return window table in pol_returns_v4.md) or [REF-2] (the refund speed table in pol_refunds.md). Since the practice questions are in English, Hindi, and Hinglish while the policy files are in English, we used sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 to embed and retrieve chunks across languages. We also wrote two small helper functions: one that looks up product details (category, price, warranty, Assured status) from products.csv, and one that calculates business-day dates (Monday to Saturday, skipping holidays.csv) from the reference date 20 October 2026.
- **Step 3 (Multi-Turn Conversation with Tools):** We turned the single-turn RAG setup into a stateful multi-turn agent connected to lookup_order and create_ticket from tools/tools.py. On Turn 1, the agent introduces itself as an AI assistant and asks for the 8-digit Order ID. On Turn 2, it looks up the order and asks for caller verification (either the last 4 digits of the customer's phone number or their 6-digit delivery pincode, following section IDV-2 of agent_identity_verification.md). If verification fails once, it allows one retry; if it fails twice, it refuses to share any order info (IDV-3). Once verified, it remembers the order across follow-up turns, answers questions about return deadlines or refund timelines, refuses out-of-policy requests (like cancelling an already shipped order or giving a Rs. 300 late voucher on standard delivery), and calls create_ticket() when a human specialist is needed.
- **Step 4 (Voice Wrapper with Gnani STT and TTS):** We wrapped the Step 3 agent with GnaniSTTClient and GnaniTTSClient (using model='timbre-v2.5' and voice='Kaveri') from the official gnani-vachana SDK, reading GNANI_API_KEY from environment variables. On the input side, we wrote a number normalizer so spoken words from STT (like 'two six two nine five zero one five') turn into digits ('26295015') before hitting the order lookup tool. On the output side, we wrote normalize_for_ear() to strip out square-bracket policy tags like [RET-2], expand ISO dates into spoken format ('22 October 2026'), and space 8-digit order IDs into two 4-digit groups ('2629 5015') so the TTS audio sounds natural.

2. Step 1 vs. Step 2 Scores (52 Practice Questions)

We evaluated both Step 1 and Step 2 on all 52 questions in practice/questions.csv against the expected gold policy sections (gold_evidence), required factual keywords (must_include), and forbidden claims (must_not_include).

Evaluation Metric (52 Questions)	Step 1: Prompt-Only	Step 2: Initial RAG	Step 2: Tuned RAG
Retrieval Hit Rate @ 4 (gold_evidence)	0/47 (0.0%)	33/47 (70.2%)	46/47 (97.9%)
Factual Keyword Match (must_include)	10/52 (19.2%)	31/52 (59.6%)	38/52 (73.1%)
Hallucination Guard (must_not_include)	52/52 (100.0%)	52/52 (100.0%)	52/52 (100.0%)
Overall Strict Pass Rate	10/52 (19.2%)	31/52 (59.6%)	38/52 (73.1%)

What we learned from comparing Step 1 and Step 2: In Step 1, the prompt-only LLM got only 10 out of 52 questions right (19.2%). It guessed generic e-commerce numbers—for example, saying ZentroMart Wallet refunds take 3 to 5 business days (actual policy in pol_refunds.md is instant / within 2 hours), claiming electronics have a 14-day return window (actual policy in pol_returns_v4.md is 7 days), and inventing a student discount on orders above Rs. 500. Adding RAG in Step 2 immediately jumped accuracy from 19.2% to 59.6% (and 73.1% after tuning section retrieval), with retrieval hitting the exact gold policy section on 46 of the 47 supported questions (97.9%).

3. Step 3 Multi-Turn Conversation Results (23 Conversations, 96 Turns)

We tested our Step 3 stateful conversation engine across all 23 practice conversations (96 customer turns) in practice/conversations.csv:

Step 3 Multi-Turn Evaluation Metric	Passed	Total Turns	Accuracy
Tool Calls Match (expected_tool_calls)	96	96	100.0%
Must-Include Keywords (must_include)	95	96	99.0%
Must-Not-Include Guard (must_not_include)	96	96	100.0%
Gold Section Retrieval Hit (gold_evidence)	96	96	100.0%

4. What Went Wrong and What We Fixed (and What We Would Fix Next)

- **Product names in questions didn't match products.csv at first:** In products.csv, product titles have color or size suffixes at the end (like 'Rangoli Cotton Straight Kurta, Blue'), while customers only ask about 'Rangoli Cotton Straight Kurta'. At first our catalog lookup missed these, so the LLM didn't know which return category the item belonged to. We fixed this by stripping the ', Color/Size' suffix when matching product names.
- **Long policy files crowded out the exact table chunk in Step 2:** In our first Step 2 run, retrieval hit rate was 70.2% because boosting an entire file like pol_refunds.md pulled sections REF-5 and REF-6 instead of REF-2 (the actual refund timeline table). Similarly, in pol_returns_v4.md, the Festive Sale section

(RET-8) was getting retrieved even when the customer didn't buy during the sale. We fixed this by adding section-level keyword boosts and only boosting RET-8 when the question mentions a sale or an October 1-15 order date, which raised retrieval hit rate from 70.2% to 97.9%.

- **Order ID digits getting confused with phone verification digits:** In Step 3, ZentroMart order IDs are 8 digits often written as two 4-digit groups (like '8978 2837'). Our regex initially mistook the second half ('2837') for the customer's 4-digit phone verification number and called lookup_order with the wrong verification value on Turn 2. We fixed this by stripping out the matched 8-digit Order ID from the text before scanning for 4-digit phone numbers or 6-digit pincodes.
- **Hindi answers getting cut off mid-sentence:** Devanagari text uses more tokens per word in the Qwen tokenizer, so max_new_tokens=96 cut off several Hindi answers mid-word. Increasing max_new_tokens to 192 fixed the truncation.
- **What we would fix next (Remaining Step 2 misses):** Even with 97.9% retrieval in Step 2, our strict keyword match was 73.1% because of three things we would fix next: (1) In our prompt we wrote 'Start every response with You are talking to an AI assistant:', and our Python loop also prepended that prefix, which caused a few questions (Q26, Q42, Q48) to fall into a repetition loop repeating the prefix; removing the duplicate instruction from the system prompt and letting Python attach the prefix cleanly fixes this. (2) For refund date questions with a range (like 5 to 7 business days for credit cards in Q06, or 1 to 3 business days for UPI in Q51), the LLM sometimes picked 3 or 5 days instead of the 7-day or 3-day upper bound expected by the grader. (3) For out-of-scope questions (like student discounts or Kochi stores), forcing a standard abstention phrase ('I do not have information on this in our documents') would pass all strict keyword checks.

5. Bonus Work

- **Multilingual & Hinglish Support without Translation Latency:** Instead of translating Hindi and Hinglish queries to English with an extra LLM call, we used a multilingual embedding model and bilingual date/policy summaries so the agent handles English, Hindi, and Hinglish directly.
- **Spoken Number Normalization for Voice Input:** Gnani STT often transcribes spoken order IDs and phone digits as words ('two six two nine...', 'eight thousand seven hundred...'). We built normalize_spoken_numbers() supporting English, Hindi, and Hinglish number words (while protecting phrases like 'last four' / 'aakhri chaar') so voice transcripts work seamlessly with tool calls.
- **Interactive Browser Microphone Loop with Auto-Pacing & Clean Call Ending:** In addition to the batch folder runner (run_batch_audio_folder), we built an interactive Colab voice loop that records from the browser microphone, calculates the exact WAV playback duration so the next recording doesn't overlap with the agent's voice, and detects when the caller says 'thank you, that is all' to deliver a warm closing message and end the call.